SG No. 165 D_{3d}^4 $P\bar{3}c1$ [trigonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `2a`, site symmetry: `32.`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `32.`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[1,2,3,-4,-5,-6]`
2	$[0,\ 0,\ -Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[7,8,9,-10,-11,-12]`

Table 2: Wyckoff bond: `6b@2a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ -X,\ Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[1,-4]`
2	$[X,\ 2X,\ Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[2,-6]`
3	$[-2X,\ -X,\ Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[3,-5]`
4	$[-X,\ X,\ -Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[7,-10]`
5	$[-X,\ -2X,\ -Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[8,-12]`
6	$[2X,\ X,\ -Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[9,-11]`

Table 3: Wyckoff bond: `6c@2a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[0,\ 0,\ \frac{1}{4}]$	`[1,5]`
2	$[0,\ X,\ 0]$	$[0,\ 0,\ \frac{1}{4}]$	`[2,4]`
3	$[-X,\ -X,\ 0]$	$[0,\ 0,\ \frac{1}{4}]$	`[3,6]`
4	$[-X,\ 0,\ 0]$	$[0,\ 0,\ \frac{3}{4}]$	`[7,11]`
5	$[0,\ -X,\ 0]$	$[0,\ 0,\ \frac{3}{4}]$	`[8,10]`
6	$[X,\ X,\ 0]$	$[0,\ 0,\ \frac{3}{4}]$	`[9,12]`

Table 4: Wyckoff bond: `12d@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[3]`
4	$[Y,\ X,\ -Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[4]`
5	$[X-Y,\ -Y,\ -Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[5]`
6	$[-X,\ -X+Y,\ -Z]$	$[0,\ 0,\ \frac{1}{4}]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[9]`
10	$[-Y,\ -X,\ Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[10]`
11	$[-X+Y,\ Y,\ Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[11]`
12	$[X,\ X-Y,\ Z]$	$[0,\ 0,\ \frac{3}{4}]$	`[12]`

* Wyckoff site: `2b`, site symmetry: `-3..`

Table 5: Wyckoff bond: `2a@2b`, bond symmetry: `-3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,2,3,-7,-8,-9]`
2	$[0,\ 0,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,6,-10,-11,-12]`

Table 6: Wyckoff bond: `6b@2b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-7]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ 0]$	`[2,-8]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ 0]$	`[3,-9]`
4	$[Y,\ X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,-10]`
5	$[X-Y,\ -Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,-11]`
6	$[-X,\ -X+Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6,-12]`

* Wyckoff site: `4c`, site symmetry: `3..`

Table 7: Wyckoff bond: `4a@4c`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2,3]`
2	$[0,\ 0,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$[0,\ 0,\ -Z]$	$[0,\ 0,\ -z]$	`[7,8,9]`
4	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10,11,12]`

Table 8: Wyckoff bond: `12b@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ z]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ z]$	`[3]`
4	$[Y,\ X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4]`
5	$[X-Y,\ -Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[5]`
6	$[-X,\ -X+Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$[0,\ 0,\ -z]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$[0,\ 0,\ -z]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$[0,\ 0,\ -z]$	`[9]`
10	$[-Y,\ -X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\ Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[12]`

* Wyckoff site: `4d`, site symmetry: `3..`

Table 9: Wyckoff bond: `4a@4d`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9]`
4	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[10,11,12]`

Table 10: Wyckoff bond: `12b@4d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3]`
4	$[Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[4]`
5	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[5]`
6	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[9]`
10	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[12]`

* Wyckoff site: `6e`, site symmetry: `-1`

Table 11: Wyckoff bond: `6a@6e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,-7]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-8]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,-9]`
4	$[Y,\ X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,-10]`
5	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,-11]`
6	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6,-12]`

* Wyckoff site: `6f`, site symmetry: `.2.`

Table 12: Wyckoff bond: `6a@6f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,-5]`
2	$[-2X,\ -X,\ Z]$	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,-4]`
3	$[X,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,-6]`
4	$[-X,\ -2X,\ -Z]$	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7,-11]`
5	$[2X,\ X,\ -Z]$	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[8,-10]`
6	$[-X,\ X,\ -Z]$	$\left[x,\ x,\ \frac{3}{4}\right]$	`[9,-12]`

Table 13: Wyckoff bond: `6b@6f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,5]`
2	$[0,\ X,\ 0]$	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,4]`
3	$[-X,\ -X,\ 0]$	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,6]`
4	$[-X,\ 0,\ 0]$	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7,11]`
5	$[0,\ -X,\ 0]$	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[8,10]`
6	$[X,\ X,\ 0]$	$\left[x,\ x,\ \frac{3}{4}\right]$	`[9,12]`

Table 14: Wyckoff bond: `12c@6f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3]`
4	$[Y,\ X,\ -Z]$	$\left[0,\ x,\ \frac{1}{4}\right]$	`[4]`
5	$[X-Y,\ -Y,\ -Z]$	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[5]`
6	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$\left[x,\ x,\ \frac{3}{4}\right]$	`[9]`
10	$[-Y,\ -X,\ Z]$	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[10]`
11	$[-X+Y,\ Y,\ Z]$	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[11]`
12	$[X,\ X-Y,\ Z]$	$\left[x,\ x,\ \frac{3}{4}\right]$	`[12]`

* Wyckoff site: `12g`, site symmetry: `1`

Table 15: Wyckoff bond: `12a@12g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[-y,\ x-y,\ z]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[-x+y,\ -x,\ z]$	`[3]`
4	$[Y,\ X,\ -Z]$	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[4]`
5	$[X-Y,\ -Y,\ -Z]$	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	`[5]`
6	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$[y,\ -x+y,\ -z]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$[x-y,\ x,\ -z]$	`[9]`
10	$[-Y,\ -X,\ Z]$	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\ Y,\ Z]$	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ Z]$	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	`[12]`